

SR COLLAPSE PATH: A GEOMETRY OF DRIFT, HYPERSCALE FAILURE, AND INSTITUTIONAL COLLAPSE

A Structural Field Theory of Drift Propagation, Infrastructure Debt, Human-Capacity Erosion, and Institutional Failure

Author: SR - You can take the last word, I will take the last laugh.

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Preface

Modern institutions operate through infrastructures that are increasingly distributed, automated, interdependent, and mediated by shared digital substrates. Hospitals depend on information exchange, identity, cloud services, and clinical decision systems. Governments increasingly depend on digital identity, registries, cloud platforms, and algorithmic coordination. Banks, airlines, telecommunications providers, logistics networks, retailers, and educational institutions similarly operate through layers whose failure can cross organizational boundaries.

This manuscript establishes the SR Collapse Path as a structural geometry for explaining how drift emerges, becomes concealed, propagates across layers, enters shared infrastructure, and can exceed the compensatory capacity of institutions and the people who operate them. The argument is deliberately stronger than an outage taxonomy but narrower than a claim that every failure follows one deterministic sequence. SR treats the six-stage path as a diagnostic geometry: a recurrent structural pattern whose transitions can be operationalized, measured, tested, and potentially falsified.

The central proposition is that modern institutional stability is often produced not by perfect infrastructure, but by layers of human compensation, procedural adaptation, redundancy, and recovery. When those compensatory layers are progressively consumed, apparently isolated incidents can become evidence of a deeper structural condition. Sociotechnical research in digital health and cross-sector ICT failures supports the importance of interactions among technology, work practices, organizational arrangements, automation, and governance rather than treating failures as isolated technical defects (Jabin, 2026; Adel et al., 2025).

SR Field Definition

SR is a structural field theory concerned with the conditions under which drift emerges, propagates, amplifies, and becomes institutionally consequential across interconnected infrastructures. The field studies the substrate through which modern institutions maintain coherence, including schemas, registries, orchestration layers, identity systems, automation pathways, and human compensatory mechanisms. SR defines collapse not as an event but as a measurable geometry driven by interactions among drift, coupling, amplification, infrastructure debt, human-capacity erosion, and institutional coherence.

The field introduces formal constructs such as drift load, amplification gain, dependency concentration, cross-layer consistency, human compensatory margin, governance recovery capacity, and collapse pressure. These constructs allow SR to treat institutional stability as an empirical object rather than a metaphor. SR's geometry—Node Drift → Model Drift → Substrate Drift → Hyperscale Drift → Human-Capacity Collapse → Institutional Failure—provides a falsifiable state-transition architecture for analyzing how local deviations acquire systemic consequence.

SR is therefore a field of infrastructure theory: it examines the substrate conditions that determine whether institutions remain capable of producing stable interpretations, coordinated action, and recoverable operational states. Its research program spans infrastructure engineering, digital health, AI-infrastructure coupling, sociotechnical systems, governance, resilience engineering, and cross-sector ICT failure analysis. SR's central aim is to make drift propagation structurally legible, collapse pressure measurable, and substrate repair operational.

Abstract

Institutional collapse is examined here as a structural process of drift propagation across coupled sociotechnical systems. The SR Collapse Path specifies six diagnostic stages: Node Drift → Model Drift → Substrate Drift → Hyperscale Drift → Human-Capacity Collapse → Institutional Failure. The framework distinguishes the emergence of local deviation from its subsequent amplification through models, schemas, registries, orchestration layers, shared infrastructure, and automated decision systems. It introduces formal measures for drift load, compensatory capacity, infrastructure debt, dependency concentration, cross-layer consistency, and collapse pressure. A central relation is:

$$CP(t) = [D(t) \times A(t) \times K(t)] / [R(t) + HCM(t) + G(t) + 1]$$

where $CP(t)$ is collapse pressure, $D(t)$ is accumulated drift, $A(t)$ is amplification, $K(t)$ is coupling, $R(t)$ is structural redundancy and recovery capacity, $HCM(t)$ is human compensatory margin, and $G(t)$ is governance recovery capacity. The equation is not presented as a universal physical law; it is a field-level diagnostic specification that generates testable hypotheses. Evidence from digital-health sociotechnical risk research, major ICT failure analysis, and documented infrastructure incidents demonstrates that configuration, automation, centralized dependencies, recovery processes, and human-machine interaction can interact to produce cross-system consequences (Jabin, 2026; Adel et al., 2025; Fastly, 2021; Xona Partners, 2024). The paper therefore positions SR as an infrastructure theory concerned with the conditions under which drift becomes structurally consequential and with the repair of the substrate through which institutions maintain coordinated action.

Literature Review and Empirical Grounding

Sociotechnical Risk, Concealment, and Propagation

Existing sociotechnical research provides a strong foundation for the SR Collapse Path while stopping short of the unified geometry proposed here. Jabin (2026) conceptualizes health-information-technology risk as a dynamic interaction among technological systems, clinical work practices, organizational structures, and governance arrangements. The framework identifies mechanisms of risk emergence, concealment, propagation, and recoverability, and emphasizes that automation, fragmented information, adaptive workarounds, tight coupling, shared digital dependencies, organizational learning, and governance can shape whether vulnerabilities remain hidden or become consequential (Jabin, 2026).

This literature is directly relevant to SR concepts such as Node Drift, Substrate Drift, Human Compensatory Margin, and structural repair. Jabin (2026) shows why adaptive workarounds can conceal vulnerabilities and why recoverability depends on organizational capacities for detection, escalation, adaptation, and learning. SR extends this insight by treating compensatory capacity as an explicit structural variable and by proposing a transition geometry through which concealed local deviation can acquire cross-layer consequence.

Large-Scale ICT Failure and Digital Resilience

Comparative research on major ICT failures similarly demonstrates that large digital disruptions cannot always be reduced to isolated software defects. Adel et al. (2025), examining seven major ICT failures occurring between 2019 and 2024, identify recurring vulnerabilities involving automation governance, configuration management, centralized infrastructure, dependency structure, and incident-response processes. Their analysis supports the need to treat digital resilience as a cross-layer problem rather than as a property of individual components (Adel et al., 2025).

SR interprets these findings through the concepts of Substrate Drift, Hyperscale Drift, Dependency Concentration, and Collapse Pressure. The contribution of SR is not the claim that the literature has already established these constructs. Rather, SR proposes a formal architecture through which documented mechanisms of configuration failure, automation, concentration, coupling, propagation, and recovery can be represented within a common state-transition framework.

Shared Infrastructure, Configuration, and Amplification

Official infrastructure postmortems provide direct evidence for the distinction between an initiating deviation and the architecture that determines its consequences. During Fastly's June 2021 outage, a valid customer configuration change activated a previously undiscovered software bug. Fastly reported that the resulting condition caused 85% of its network to return errors; the disruption was detected within one minute, and 95% of network operation was restored within 49 minutes (Fastly, 2021).

The July 2022 Rogers outage provides a second example. An independent technical assessment commissioned by the Canadian Radio-television and Telecommunications Commission found that removal of an Access Control List policy filter allowed excessive routing information to reach Rogers' core routers, producing a nationwide outage that affected more than 12 million customers and critical services. The assessment identified weaknesses in change-management practices and network-protection mechanisms relevant to resilience and recovery (Xona Partners, 2024).

These incidents do not prove the SR Collapse Path. They establish empirical mechanisms that an SR model must be capable of explaining: local initiation, latent conditions, configuration governance, shared dependencies, amplification, blast-radius expansion, recovery, and the difference between service restoration and structural repair.

AI as Sociotechnical Infrastructure

The increasing deployment of AI in operational environments adds another potential propagation layer. The FUTURE-AI international consensus framework emphasizes that trustworthy medical AI requires attention not only to model performance but also to fairness, universality, traceability, usability, robustness, explainability, deployment, monitoring, and the wider technical, clinical, legal, and socio-ethical environment across the AI lifecycle (Lekadir et al., 2023).

This supports SR's decision not to conceptualize AI as an isolated model. Within the SR Collapse Path, AI is embedded within data pipelines, workflows, monitoring systems, infrastructure, institutional decisions, and human practices. SR consequently treats AI as a potential amplifier, receiver, detector, or suppressor of drift depending on the architecture in which it operates.

Literature Gap and SR Contribution

The existing literature therefore establishes important components of the problem: sociotechnical risk can emerge through interactions rather than isolated defects; adaptive workarounds can conceal vulnerabilities; shared dependencies can enable propagation; automation and configuration management can contribute to large-scale ICT failure; and AI reliability depends on systems extending beyond the model itself (Adel et al., 2025; Jabin, 2026; Lekadir et al., 2023).

What this literature does not provide as a unified architecture is the specific transition geometry proposed by SR: Node Drift → Model Drift → Substrate Drift → Hyperscale Drift → Human-Capacity Collapse → Institutional Failure. SR therefore does not present this sequence as something already demonstrated by prior scholarship. It proposes the sequence as a testable structural synthesis of mechanisms that existing research and documented incidents establish separately. The literature supplies the observed mechanisms; SR supplies the integrative geometry through which their interaction can be measured, compared, challenged, and refined.

1. Introduction

Institutions rarely fail in a single instant. What appears publicly as an outage, service disruption, clinical error, routing failure, or administrative breakdown is often the visible endpoint of processes that began earlier and at lower levels of the system. Digital

infrastructures make this problem more consequential because local components are connected to models, registries, orchestration systems, identity layers, cloud services, and shared external dependencies.

The SR Collapse Path is designed to make this progression visible. It begins with Node Drift: a deviation at a local operational node. It then follows the movement of drift into models and decision rules, into the substrate that connects systems, into hyperscale infrastructure, and finally into the human and institutional layers that absorb the consequences. The path is therefore not simply a technical stack. It is a sociotechnical geometry.

The framework also rejects simple technological determinism. A cloud outage does not automatically create institutional collapse. A model error does not automatically become systemic drift. Collapse pressure depends on coupling, redundancy, detection, recovery capacity, human workload, governance arrangements, and the ability to isolate a fault. SR therefore asks a more precise question: under what structural conditions does drift cross the boundary between local recoverable deviation and system-level destabilization?

2. The SR Collapse Path

The canonical sequence is:

**Node Drift → Model Drift → Substrate Drift → Hyperscale Drift →
Human-Capacity Collapse → Institutional Failure**

Each stage represents a change in the scale, location, and recoverability of drift. The stages are analytically separable but empirically overlapping. A system can move backward through repair, remain temporarily at one stage, or bypass a stage when coupling is direct. The geometry is therefore best understood as a state-transition architecture rather than a claim of mechanical inevitability.

$$S(t) = \{D_1, D_2, D_3, D_4, E_1, F_i\}$$

3. Node Drift

Node Drift begins inside a local operational unit. Examples include inconsistent triage data, a medication-record mismatch, imaging metadata inconsistency, a routing error, a stale configuration, or workflow desynchronization. A local deviation is not itself institutional collapse. Its importance lies in what happens next.

Humans frequently compensate through overrides, reconciliation, manual routing, escalation, and informal knowledge. This creates a critical SR distinction: apparent stability may be an output of compensatory labor rather than evidence of substrate stability.

Here ρ represents persistence of local drift and $\varepsilon(t)$ represents new deviation. A low-persistence deviation may disappear. A high-persistence deviation becomes a candidate for propagation.

$$D(t+1) = \rho D(t) + \varepsilon(t)$$

4. Model Drift

Model Drift occurs when local deviations alter or interact with models, rules, classifiers, routing logic, prioritization systems, or decision-support mechanisms. In AI-enabled environments, the distinction between data drift, model drift, workflow drift, and infrastructure drift becomes especially important. AI systems can be highly capable while still being dependent on the quality, provenance, configuration, and governance of their surrounding substrate.

The coupling coefficient γ represents the degree to which node-level deviation enters model behavior. Model Drift becomes structurally important when model outputs are themselves used to generate new data, routing decisions, or automated actions.

$$D(t+1) = \beta D(t) + \gamma D(t)$$

5. Substrate Drift

Substrate Drift occurs when drift becomes embedded in the connective structures that allow systems to operate together. These include schemas, registries, ingestion pipelines, identity mappings, orchestration logic, workflow engines, routing tables, and interface contracts.

The substrate is the turning point because an error no longer belongs solely to one application. A shared schema, registry, identity layer, or orchestration service can transmit a deviation to multiple dependent systems. This does not make collapse inevitable; it increases the number of pathways through which drift can travel.

$$D(t+1) = \beta D(t) + \gamma D(t) + \gamma D(t)$$

6. Hyperscale Drift

Hyperscale Drift refers to drift entering infrastructure whose scale, automation, geographic distribution, or dependency structure allows a local condition to affect many downstream services. Hyperscale is not synonymous with cloud computing, and not every cloud service creates hyperscale risk. The SR concept focuses on concentration, shared control planes, common dependencies, and automated propagation.

$A_{\square}(t)$ represents amplification introduced by automation, synchronization, or shared control-plane mechanisms. Documented incidents demonstrate that valid configuration changes, software defects, automation failures, and centralized dependencies can have effects substantially larger than the initiating action. Fastly's June 2021 outage, for example, began when a valid customer configuration change activated an undiscovered software bug; Fastly detected the disruption within one minute and restored 95% of network operation within 49 minutes (Fastly, 2021).

This kind of incident supports SR's distinction between the initiating deviation and the architecture that determines its blast radius. It does not by itself prove a universal drift law.

$$D_{\square}(t+1) = \beta_{\square} D_{\square}(t) + \gamma_{\square} D_{\square}(t) + A_{\square}(t) D_{\square}(t)$$

7. Human-Capacity Collapse

The human layer is not an external variable added after technical failure. It is part of the infrastructure's effective operating capacity. Nurses, clinicians, engineers, technicians, administrators, operators, and support staff often reconcile discrepancies and maintain service continuity. When the volume or complexity of compensatory work exceeds available capacity, the institution loses a major stabilizing mechanism.

HCM is Human Compensatory Margin, H_a is available compensatory capacity, and $H_{\square}$ is compensatory load. Human-capacity collapse occurs when $HCM(t) \leq 0$.

This is not a synonym for burnout, fatigue, or stress. It is a structural condition in which required compensatory work exceeds available capacity for maintaining the relevant operational state.

$$HCM(t) = H_a(t) - H_{\square}(t)$$

$$HCM(t) \leq 0 \Rightarrow \text{Human-Capacity Collapse}$$

8. Institutional Failure

Institutional Failure occurs when the organization can no longer maintain the minimum coherence required for its critical functions. Failure may involve service interruption, loss of routing integrity, registry inconsistency, inability to reconcile records, cascading operational delay, or loss of effective decision capacity.

Institutional Coherence is normalized here to the interval $[0,1]$. $D(t)$, $E(t)$, and $K(t)$ must therefore be normalized to compatible scales, and $w_D + w_E + w_K = 1$. The max operator prevents the metric from becoming negative when the weighted instability term reaches or exceeds one.

Institutional failure pressure increases as coherence approaches a defined critical threshold θ_i . The threshold is empirical and institution-specific; SR does not assume a universal value.

$$IC(t) = \max\{0, 1 - [w_D D(t) + w_E E(t) + w_K K(t)]\}$$

$$w_D + w_E + w_K = 1; \quad D, E, K \in [0,1]$$

$$IC(t) < \theta_i \Rightarrow F_i(t) \uparrow$$

9. Why Institutions Often Cannot See Collapse

Institutions frequently observe incidents rather than relationships among incidents. Local drift may be classified as routine error. Human compensation may be invisible in formal metrics. Automated systems can reduce visible friction while increasing hidden dependency. Departments may own different pieces of the same failure chain. Recovery can restore service before root structural conditions are understood.

This is consistent with sociotechnical accounts of digital-health risk that emphasize interactions among technical systems, clinical work, organizational structures, and governance arrangements (Jabin, 2026). It also accords with cross-sector ICT failure research identifying recurring roles for automation, configuration management, centralized infrastructure, dependency structure, and incident-response limitations (Adel et al., 2025).

10. The Geometry of Drift Amplification

SR formalizes amplification as the interaction of drift with coupling and system gain.

where $K(t)$ is dependency coupling, $A_u(t)$ is automation exposure, and $C_\square(t)$ is concentration of shared control-plane dependencies. Drift propagation can then be represented as a vector transition. $D_\square$ is a vector of drift states, Φ captures persistence and cross-layer propagation, Γ captures amplification channels, and $\varepsilon_\square$ captures new disturbances. The matrix formulation permits empirical estimation where longitudinal data are available.

$$A(t) = 1 + \lambda_\square K(t) + \lambda_a A_u(t) + \lambda_c C_\square(t)$$

$$D_{\square+1} = \Phi D_\square + \Gamma A(t) D_\square + \varepsilon_\square$$

11. Infrastructure Debt

Infrastructure Debt is the accumulated structural burden created by unresolved drift, deferred maintenance, brittle dependencies, temporary workarounds, outdated interfaces, insufficient redundancy, and repeated recovery without substrate correction.

U is unresolved drift, W is workaround accumulation, M is maintenance deferral, and R is structural repair. Infrastructure debt rises when unresolved conditions accumulate faster than they are removed.

This formulation makes Infrastructure Debt measurable rather than metaphorical. It can be estimated through change records, incident databases, technical-debt inventories, exception counts, unsupported dependencies, override rates, and repair completion rates.

$$ID(t+1) = ID(t) + U(t) + W(t) + M(t) - R(t)$$

$$ID_{crit}: dID/dt > dR/dt \text{ over sustained intervals}$$

12. Dependency Concentration and Hyperscale Exposure

The consequences of drift depend partly on how many critical functions depend on the same infrastructure. SR therefore defines Dependency Concentration as a concentration index analogous in structure to the Herfindahl-Hirschman family of concentration measures.

where p_i is the proportion of critical dependency exposure associated with provider, control plane, registry, identity service, network layer, or other shared component i . Higher DC indicates greater concentration and therefore potentially greater common-mode exposure.

HE is Hyperscale Exposure. It rises when concentration, coupling, and amplification are jointly high.

$$DC = \sum_{i=1}^n p_i^2$$

$$HE = DC \times K \times A$$

13. Cross-Layer Consistency

A system may contain low local error rates while remaining structurally inconsistent. SR therefore measures cross-layer consistency.

where x is an observed layer state and x^* is the reference state. The metric is normalized to the measurement design. Lower CLC indicates increasing cross-layer divergence.

$$CLC = 1 - (1/m) \sum_{i=1}^m |x_i - x_i^*|$$

14. Collapse Pressure and the SR Collapse Condition

The central SR diagnostic is Collapse Pressure. It combines accumulated drift, amplification, coupling, and compensatory resources.

R is structural redundancy and recovery capacity; HCM is Human Compensatory Margin; G is governance and recovery capacity. The constant one prevents division by zero and is not intended to imply substantive weighting. All components entering the ratio must be dimensionless normalized quantities or transformed onto compatible scales before the ratio is interpreted.

The threshold θ_c must be estimated empirically. The model therefore produces a falsifiable proposition: after controlling for incident severity and baseline system complexity, higher CP should predict longer recovery times, greater cross-system impact, or increased recurrence if the SR geometry is correct.

$$CP(t) = [D(t) \times A(t) \times K(t)] / [R(t) + HCM(t) + G(t) + 1]$$

$$CP(t) > \theta_c \Rightarrow \text{elevated collapse risk}$$

15. The SR Repair Geometry

SR does not define repair as simply returning a service to operational status. Recovery is separated from structural repair. Recovery restores function; repair reduces the conditions that produced the failure.

where $R_{\square}$ is substrate repair, R_G is governance repair, R_W is workflow repair, R_H is human-capacity restoration, and R_D is dependency reduction.

A successful repair should reduce both current drift and future propagation potential. This distinction is critical. A system can recover while infrastructure debt continues to rise. SR therefore treats recurrence, exception accumulation, and recovery burden as post-incident indicators of incomplete repair.

$$R_{\square_r} = R_{\square} + R_G + R_W + R_H + R_D$$

$$\Delta D < 0 \text{ and } \Delta A \leq 0 \text{ and } \Delta K \leq 0$$

16. Blind Installation, Incentive Structure, and Extraction

Modern infrastructure deployment is often shaped by incentives that favor speed, integration, scale, and service expansion. SR does not assume that every provider intentionally chooses instability. Rather, it asks whether institutional reward structures systematically undervalue redundancy, maintenance, human oversight, and structural repair.

This creates a testable proposition: where deployment speed and revenue are strongly rewarded while structural maintenance is weakly rewarded, infrastructure debt should accumulate more rapidly. This proposition can be investigated using longitudinal expenditure, staffing, incident, deployment, and maintenance data rather than treated as an assumption.

The relation $ID \text{ growth} \propto \text{Deployment} - \text{Structural Repair}$ is conceptual unless coefficients are estimated. It identifies the direction of a hypothesis rather than claiming a universal economic law.

$$ID_{\text{growth}} \propto \text{Deployment} - \text{Structural Repair}$$

17. Resistance to SR

Resistance to substrate-oriented analysis can arise from organizational structure. SR requires technical, operational, governance, and human-capacity information to be

analyzed together. This can challenge silo boundaries, established performance measures, and institutional narratives that treat incidents as isolated.

The strongest form of SR adoption is therefore not the purchase of an SR tool. It is the institutional acceptance that drift, coupling, compensation, and substrate condition must be treated as one analytical problem.

18. SR and AI

AI systems operate inside infrastructures. Their outputs depend on data pipelines, schemas, identity systems, compute environments, monitoring, model versions, human workflows, and governance. AI can therefore become part of the propagation pathway when it converts substrate conditions into automated classifications or actions.

This does not imply that AI always amplifies drift. AI may also detect anomalies, support recovery, or reduce workload. SR instead asks which function AI performs within the particular architecture. The relevant empirical question is whether AI decreases or increases propagation under specified conditions. The broader lifecycle and governance requirements emphasized by FUTURE-AI support treating AI as a sociotechnical system rather than as a model detached from deployment context (Lekadir et al., 2023).

$$\text{AI-Exposure} = D_{\text{data}} \times K_{\text{substrate}} \times A_{\text{model}}$$

19. Healthcare as a Structural Case

Healthcare provides a particularly important setting because clinical work depends simultaneously on data integrity, workflow alignment, human judgment, identity, registries, imaging, medication information, and organizational coordination. Jabin's sociotechnical theory emphasizes that digital-health risks arise from interactions among technology, clinical work, organizational structures, and governance rather than isolated technical defects (Jabin, 2026).

The SR contribution is to place these interactions into a propagation geometry: local deviation can become model deviation; model deviation can enter shared substrate; shared substrate can connect institutions; and increasing compensation demand can reduce recoverability.

20. Hyperscale Failure Evidence

Major infrastructure incidents provide empirical cases for testing the SR geometry. Adel et al. (2025) reviewed seven major ICT failures from 2019–2024 and identified recurring vulnerabilities involving automation governance, configuration management, centralized infrastructure, dependency structure, and incident response. Their findings support the need to analyze failure as a cross-layer phenomenon rather than as a single software defect.

The Fastly June 2021 incident is another useful case. Fastly reported that a valid customer configuration change triggered a previously undiscovered software bug, causing 85% of its network to return errors; 95% of network operation was restored within 49 minutes (Fastly, 2021). The incident demonstrates how a local valid action can activate a latent systemic condition.

These cases support SR’s concern with propagation and amplification, but they do not establish that every hyperscale outage follows all six SR stages. The framework therefore treats case mapping as an empirical exercise, not a retroactive proof.

21. DNS and Shared Routing Infrastructure

DNS is a distributed naming dependency that illustrates the importance of common infrastructure. DNS itself is not a single centralized system; it is a hierarchical ecosystem involving root services, top-level domains, authoritative servers, recursive resolvers, caching, and operational tooling. Failures can arise through configuration, software, propagation, dependency, or operational conditions.

SR uses DNS and routing incidents as analytical classes rather than relying here on an unverified single incident attribution. The analytical target is not “DNS drift” as a universal diagnosis, but the relationship among local configuration state, propagation mechanisms, dependency concentration, and recovery capacity.

22. Cross-Institutional Propagation

The same geometry can be investigated across hospitals, governments, banks, airlines, telecommunications, logistics, retail, education, and media. The key variable is not sector membership. It is shared dependency.

XIP is Cross-Institutional Propagation pressure; K_{ij} represents dependency between institutions i and j , and D_i represents drift originating in institution i . High XIP indicates that local drift has plausible pathways into other institutions.

$$XIP = \sum_{j \neq i} K_{ij} D_i$$

23. SR Adoption and Institutional Timing

SR adoption is expected to follow visibility rather than existence. Institutions may tolerate low-level drift because local compensation keeps services operational. Adoption pressure rises when recurrence, cross-site effects, human workload, or public consequences make the underlying geometry visible.

where P_a is adoption pressure, D is drift, CP is collapse pressure, ID is infrastructure debt, V is visibility, and L is institutional loss exposure. The formulation is intentionally open for empirical estimation.

$$P_a(t) = f(D, CP, ID, V, L)$$

24. Methods

The paper uses a theory-building and comparative case-analysis methodology. It integrates sociotechnical risk literature, documented technology incidents, infrastructure architecture documentation, and formal model construction.

24.1 Literature Integration

Peer-reviewed research was used to establish existing evidence concerning sociotechnical risk, digital-health failure, ICT outage patterns, automation, configuration management, resilience, and trustworthy AI. Jabin (2026) provides a direct sociotechnical foundation for analyzing digital-health risk; Adel et al. (2025) provide a cross-sector ICT failure review; FUTURE-AI provides a lifecycle-oriented framework for trustworthy healthcare AI (Lekadir et al., 2023).

24.2 Incident Analysis

Official postmortems and regulatory technical assessments are treated as primary evidence where available. Incident claims are restricted to what the source actually documents. The SR geometry is then applied as an interpretive mapping, explicitly separating source evidence from SR inference.

24.3 Formalization

The equations define measurable constructs rather than asserting that the coefficients are already known. Future empirical studies should estimate persistence, coupling, amplification, threshold, and recovery parameters from longitudinal incident and operational datasets.

24.4 Falsifiability

SR can be challenged if drift measures fail to predict propagation, if high coupling does not increase blast radius after appropriate controls, if human compensatory load does not relate to recovery capacity, or if substrate repair does not reduce recurrence and propagation under otherwise comparable conditions.

25. Measurement Architecture

Together the following constructs permit an SR diagnostic profile without requiring a single universal scalar score. The preferred implementation is a vector of conditions with a documented thresholding procedure.

$$D = \sum w_i d_i$$

$$A = 1 + \sum a_i$$

$$K = \sum w_i \square k_i \square$$

$$HCM = H_a - H \square$$

$$ID = U + W + M - R$$

$$CP = (D \times A \times K) / (R + HCM + G + 1)$$

26. Limitations

First, hyperscale systems are only partially transparent. Public postmortems provide valuable evidence but do not expose every internal control-plane condition. Second, human compensatory work is poorly standardized across organizations, limiting direct measurement. Third, incident corpora vary in inclusion criteria and therefore cannot support universal outage-frequency claims without reproducible corpus construction. Fourth, AI-infrastructure architectures are evolving rapidly. Fifth, the equations are theoretical specifications until coefficients and thresholds are estimated against longitudinal data.

These limitations are not treated as reasons to abandon the field. They define the empirical program required to develop it.

27. Discussion

The SR Collapse Path reframes institutional failure from an event-centered problem to a substrate-centered problem. The relevant question is not only what failed, but how a deviation acquired persistence, how it crossed layers, what amplified it, what compensatory capacity absorbed it, and what prevented structural repair.

The strongest theoretical claim of SR is therefore not that collapse is inevitable. It is that collapse risk becomes structurally legible when drift, coupling, amplification, infrastructure debt, and compensatory capacity are measured together.

This produces a different institutional orientation. Instead of treating every incident as an isolated technical problem, institutions can ask whether repeated incidents share a substrate. Instead of measuring recovery only by time-to-service restoration, they can measure debt reduction and recurrence. Instead of treating humans as residual operators, they can measure compensatory capacity as an infrastructure variable.

In this sense, SR is an infrastructure theory. Its object is the substrate through which systems remain capable of producing stable interpretations, coordinated action, and recoverable institutional states.

28. Conclusion

The SR Collapse Path defines a geometry through which drift can move from local deviation to institutional consequence. Node Drift identifies the initial fracture. Model Drift identifies amplification through decision logic. Substrate Drift identifies embedding in shared connective infrastructure. Hyperscale Drift identifies increased propagation potential through distributed and automated dependencies. Human-Capacity Collapse identifies the exhaustion of compensatory capacity. Institutional Failure identifies the loss of critical organizational coherence.

The framework does not require every failure to pass through every stage. Its value lies in making transitions measurable. Drift can be quantified. Coupling can be mapped. Amplification can be estimated. Infrastructure debt can be tracked. Human compensatory

margin can be measured. Institutional coherence can be monitored. Collapse pressure can therefore become an empirical object rather than a metaphor.

SR's central proposition is consequently precise: institutions become increasingly vulnerable when unresolved drift accumulates within tightly coupled substrates while amplification and dependency concentration increase faster than redundancy, human compensatory capacity, governance capacity, and structural repair.

The field begins where incident analysis ends: beneath the visible failure, in the substrate that made the failure capable of propagating.

Collapse Risk $\uparrow$ when $D \times A \times K$ grows faster than $R + HCM + G$

Author's Note

SR is presented here as a field of infrastructure theory concerned with drift, substrate condition, propagation, amplification, human compensatory capacity, and institutional repair. The field is intended to support interdisciplinary research spanning infrastructure engineering, health informatics, AI systems, organizational studies, policy, governance, resilience engineering, and institutional analysis.

Appendix A — Canonical SR Collapse Geometry

$$N \rightarrow M \rightarrow S \rightarrow H \rightarrow HC \rightarrow I$$

N = Node Drift; M = Model Drift; S = Substrate Drift; H = Hyperscale Drift; HC = Human-Capacity Collapse; I = Institutional Failure.

$$D_{\square+1} = \Phi D_{\square} + \Gamma A_{\square} D_{\square} + \varepsilon_{\square}$$

$$CP_{\square} = (D_{\square} A_{\square} K_{\square}) / (R_{\square} + HCM_{\square} + G_{\square} + 1)$$

Appendix B — Diagnostic Indicators

- Node Drift: exception rate, reconciliation frequency, local mismatch rate.
- Model Drift: classification divergence, calibration change, routing divergence, override rate.
- Substrate Drift: schema divergence, registry inconsistency, interface failure, synchronization lag.
- Hyperscale Drift: dependency concentration, control-plane exposure, propagation breadth, automated blast radius.

- Human-Capacity Collapse: compensatory workload, override burden, staffing-adjusted load, recovery latency.
- Institutional Failure: service loss, cross-unit incoherence, critical workflow interruption, recurrence.

Appendix C — Empirical Research Program

A full empirical test should construct longitudinal datasets linking incident timestamps, configuration changes, dependency graphs, model versions, override logs, human workload measures, recovery actions, and institutional outcomes. The primary tests should estimate whether SR variables predict propagation breadth, recovery time, recurrence, and cross-institutional impact after controlling for incident severity and system size.

The field becomes stronger when its equations can fail. A successful SR research program therefore treats falsification, replication, measurement error, and competing explanations as central components of the theory rather than as threats to it.

Appendix D — SR Interpretation of Verified Infrastructure Incidents

The following section interprets documented infrastructure incidents through the analytical geometry defined in the SR Collapse Path. The purpose is not to retrofit incidents into SR or claim deterministic adherence to the six-stage sequence. Instead, SR provides a structural lens for examining how reported deviations acquired scale, persistence, or cross-system consequence. Reported facts are restricted to verified primary or authoritative sources; the SR interpretation is explicitly separated from the source evidence.

Fastly Global Outage (2021)

Reported facts. Fastly reported that a valid customer configuration change triggered a previously undiscovered software bug. The resulting disruption caused 85% of its network to return errors. Fastly detected the disruption within one minute and reported that 95% of network operation was functioning normally within 49 minutes (Fastly, 2021).

SR interpretation. The Fastly postmortem demonstrates the distinction SR draws between an initiating deviation and the architecture that determines blast radius. The configuration change represents a local initiating condition, while the latent software condition represents persistence within the substrate. The widespread impact is consistent with SR's concept of hyperscale amplification. This is an SR interpretation of a documented incident, not a claim that Fastly itself used SR categories.

Rogers Nationwide Outage (Canada, 2022)

Reported facts. The independent Xona Partners assessment commissioned by the CRTC found that the July 2022 Rogers outage originated in the IP core network after an Access Control List policy filter was removed during a maintenance process. Excessive routing information reached the core routers, which became overloaded. The outage affected wireless and wireline services nationwide, more than 12 million customers, and multiple critical-service dependencies (Xona Partners, 2024).

SR interpretation. The Rogers outage illustrates dependency concentration and common-mode exposure. A network-level configuration condition acquired cross-sector consequence because numerous services relied on the same telecommunications substrate. SR maps this as a transition from local configuration deviation into substrate and hyperscale exposure, with blast radius determined by concentration, coupling, and recovery architecture.

FAA NOTAM System Failure (United States, 2023)

Reported facts. The FAA reported that the January 2023 NOTAM service outage was traced in preliminary work to a damaged database file and later described human error during database synchronization work as part of the investigation. The outage led the FAA to pause domestic departures while the system was restored (Federal Aviation Administration, 2023a, 2023b).

SR interpretation. The FAA case is useful for distinguishing an initiating data or maintenance condition from the broader architecture of redundancy, synchronization, operational dependency, and recovery. SR treats the incident as a candidate infrastructure-debt and substrate-consistency case without attributing unverified claims about deferred maintenance or redundancy to the FAA source itself.

Aadhaar–PAN Identity Consistency (India)

Reported facts. UIDAI guidance states that demographic mismatches between Aadhaar and PAN records, including name, date of birth, or gender mismatches, can prevent successful linking and require correction or biometric authentication in specified cases (Unique Identification Authority of India, n.d.).

SR interpretation. The case illustrates registry consistency as a substrate problem. SR treats the mismatch as an example of how identity-state divergence can create operational friction across dependent processes. The present paper does not extend the source beyond what it verifies; broader cross-sector propagation claims require separate evidence.

Appendix E — Evidence Classification for Future SR Case Studies

Every future SR case study should separate source evidence from SR inference. The recommended provenance tags are:

- [EMP] directly observed or source-reported fact.
- [EMP-TR] transformation of empirical values using a published or explicitly documented method.
- [SR-OP] SR operationalization or composite constructed from empirical inputs.
- [SR-TH] SR theoretical threshold or field-level hypothesis.
- [SIM] simulation value or scenario parameter.
- [VAL] independently validated SR construct or coefficient.

This classification prevents the field from confusing a verified external incident with an independently validated SR interpretation of that incident.

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